Preferments: The Foundations

What a preferment is, and the two axes that sort them: microbiology and hydration

Sourdough Science

github.com/zrrbite/sourdough-science

What this deck covers

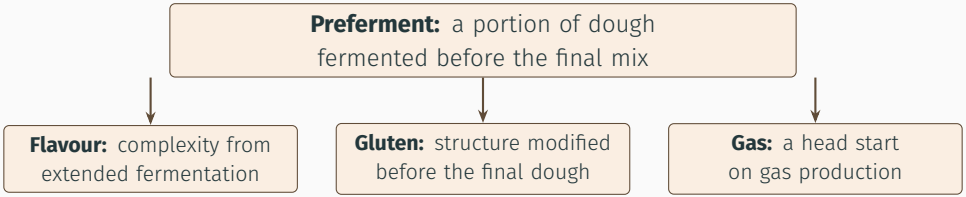
Practical questions, answered from Part I of the notes:

- What does a preferment actually do for my dough?
- Biga, poolish, stiff levain, liquid levain: what separates them?
- What does a wild culture do that commercial yeast cannot?
- How do I steer a levain toward a sharper or a milder sour?
- Why would a stiff levain survive a hot build better than a liquid one?

Source: chapters 01–04 of the notes

A preferment and its two axes

A preferment is dough fermented ahead of the final mix



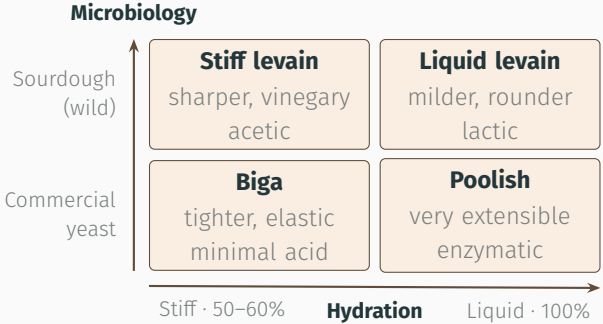
- Every preferment does these three jobs
- They diverge sharply in microbiology and hydration, and so in how the dough behaves and what the bread becomes

In the kitchen: choosing a preferment is choosing its organisms and its water; those two decide what it does to your dough and your bread.

Source: ch. 01, *What Is a Preferment?*

Two independent choices define every preferment

- **Microbiology:** a monoculture of commercial *S. cerevisiae*, or a mixed consortium of wild yeast and lactic acid bacteria (LAB)
- **Hydration:** stiff (50–60%) or liquid (100%)
- Four types, each with its own flavour, fermentation kinetics and dough rheology



In the kitchen: make the two decisions separately: which organisms, and how much water.

Source: ch. 02, *The Four Quadrants*

Each quadrant has its own flavour and dough

	Stiff (50–60%)	**Liquid (100%)**
Commercial yeast	**Biga** · Italian tradition Long, cool ferment (12–24 h) Yeast aromatics: esters, aldehydes, alcohols Minimal acidification Tighter, more elastic gluten	**Poolish** · French/Polish origin 8–16 h at room temperature More amylase and protease (more water) Sweeter, wheaty flavour Very extensible dough
Sourdough (wild)	**Stiff levain** · heterofermentative LAB Lower water activity: toward acetic acid Sharper, more vinegary tang Wider timing window for the baker	**Liquid levain** · homofermentative LAB Higher water activity: promotes lactate Milder, rounder acidity Handles like a poolish; full sourdough microbiology

In the kitchen: extensible dough: poolish. Elastic gluten: biga. Wider timing window: stiff levain.

Source: ch. 02, *The Four Quadrants*

One yeast or a consortium

Biga and poolish run on one yeast: aroma, not acid

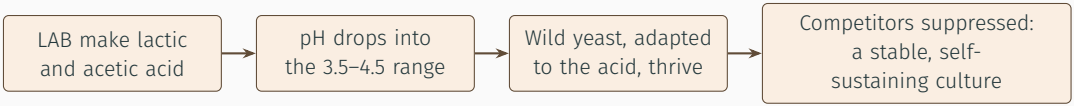
- A pure culture of *S. cerevisiae*: domesticated, single strain
- The long ferment is not there to grow a microbiome; it gives the yeast time to build flavour precursors slowly
- Alcohols, esters and aldehydes accumulate: nuanced aroma without significant acidification
- Some minor organic acid from the yeast's own metabolism, nothing near the lactic and acetic acid that LAB make
- The dough stays near neutral (pH 5.5–6.0), so acid does not modify the gluten

In the kitchen: a long biga or poolish buys aroma, not sourness. For a sweet, wheaty character from enzymatic starch breakdown, the wetter poolish is the stronger of the two.

Source: ch. 03, *Microbiology*; ch. 04, *Acidity*

A levain is a stable partnership of wild yeast and bacteria

- **Wild yeast:** mostly wild *S. cerevisiae* at home, often *Kazachstania humilis* in bakeries
- **Lactic acid bacteria (LAB):** mostly lactobacilli, such as *L. plantarum*, *L. brevis*, *F. sanfranciscensis*
- A stable association; how symbiotic is debated (Starters deck)



In the kitchen: the sourness is not a side effect. The low pH is what keeps competing organisms out and the culture stable.

Source: ch. 03, *Microbiology*; Landis et al. (2021); Carbonetto et al. (2020)

The mixed culture changes the dough in ways yeast alone cannot

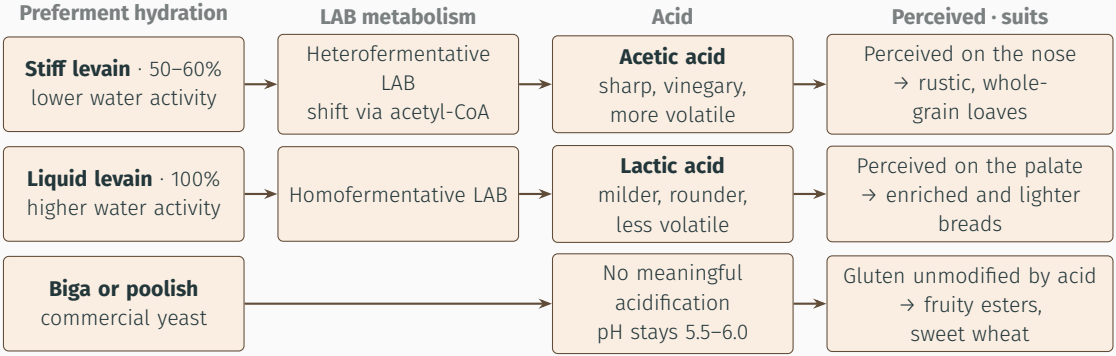
- **Gluten is cut:** proteolytic enzymes from the LAB partially degrade the gluten network during fermentation
- **Gas is uneven:** organisms with different growth curves and temperature optima compete and cooperate, so gas production is less uniform
- **Gluten charge shifts:** the acid load directly modifies gluten charge state and hydration behaviour
- **contested** Bottled acid is not the same: several studies report that adding acid directly *firms* dough and cuts extensibility, while long fermentation (acid plus time-dependent proteolysis) softens it. The notes describe the fermentation case.

In the kitchen: expect a levain dough to keep changing as it ferments, in ways biga and poolish cannot reproduce.

Source: ch. 03, *Microbiology*; charge-state mechanism: Clarke et al. (2004); contested note: ch. 25

Acid and hydration

Hydration steers which acid a levain makes



In the kitchen: hydration is one of the most important levers you have on the type and intensity of a levain's acid.

Source: ch. 04, *Acidity: Lactic vs. Acetic and the Hydration Effect*

Acetic reads on the nose, lactic on the palate

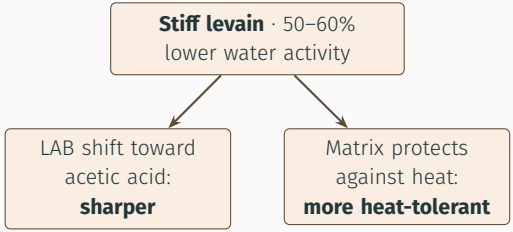
	Stiff levain · 50–60% · lower water activity	**Liquid levain** · 100% · higher water activity
LAB metabolism	Heterofermentative; shifts toward acetic acid via acetyl-CoA pathways when water is limiting	Homofermentative; produces primarily lactic acid
Character	Sharper, more vinegary tang; more volatile, so perceived more intensely by the nose	Milder, rounder; less volatile, so perceived mainly on the palate
Suits	Hearty, rustic loaves with whole-grain or high-extraction flour, where the sharpness reads as complexity	Enriched doughs, lighter breads, or wherever the acid should stay in the background and let wheat, butter or other ingredients come forward

In the kitchen: rustic whole-grain loaf: build the levain stiff. Enriched or lighter bread: liquid.

Source: ch. 04, *Acidity: Lactic vs. Acetic and the Hydration Effect*

The stiff levain is also the heat-tolerant one

- Lower water activity does more than shift LAB toward acetic acid
- It makes the matrix itself protective against heat
- Why: irreversible protein unfolding and membrane disordering are both water-mediated
- Matters only for hot builds (ch. 11)



In the kitchen: building levains hot? A stiff one gives the culture more protection, and the sharper acid comes with it.

Source: ch. 04, *Acidity*; detail in ch. 11, *Water Activity: The Matrix Is a Thermal Protectant*

Takeaways

- A preferment is dough fermented ahead: flavour, gluten change and a head start on gas
- Two independent choices define it: commercial yeast or wild culture; stiff (50–60%) or liquid (100%)
- Biga and poolish: one yeast, aroma without sourness; the dough stays at pH 5.5–6.0
- A levain: wild yeast plus LAB, which drop the pH to 3.5–4.5; acid and LAB enzymes reshape the gluten
- Stiff levain → acetic: sharp, on the nose, for rustic whole-grain loaves
- Liquid levain → lactic: mild, on the palate, for enriched and lighter breads
- The stiff levain is also the more heat-tolerant one, if you build hot

Sources

- **Clarke CI et al. (2004)** Wheat sourdough fermentation: effects of time and acidification on fundamental rheological properties. *Cereal Chemistry* 81(3):409–417. Source for the charge-state mechanism.
- **Thiele C, Grassl S, Gänzle MG (2004)** Gluten hydrolysis and depolymerization during sourdough fermentation. *J. Agric. Food Chem.* 52(5):1307–1314.
- **Wehrle K, Grau H, Arendt EK (1997)** Effects of lactic acid, acetic acid, and table salt on fundamental rheological properties of wheat dough. *Cereal Chemistry* 74(6):739–744.
- **De Vuyst L, Van Kerrebroeck S, Leroy F (2017)** Microbial ecology and process technology of sourdough fermentation. *Advances in Applied Microbiology* 100:49–160.
- **Carbonetto B et al. (2020)** Interactions between *Kazachstania humilis* yeast species and lactic acid bacteria in sourdough. *Microorganisms* 8(2):240. No mutual benefit in population terms; the pair still fermented better.
- **Landis EA et al. (2021)** The diversity and function of sourdough starter microbiomes. *eLife* 10:e61644. Who lives in 500 home starters.

Contested evidence, ch. 25: chemical and biological acidification do not produce equivalent rheology.